

Weekly Engineering Assignment 2 - Group 3 - S23

Rules Engine for Safety Hazards

Section 1: Risk Calculation

This program determines risk rating based on the probability and risk of an event. In the process, you can choose a severity scale and also get a related explanation.

Step 1: Hazard Probability

1. To simplify the code, write a function for the description part separately. Also, this function returns ov.
2. The number entered is classified through an if-else statement.
3. The correct information is printed according to the classification.

```
In [18]: a = int(input(print("Enter the probability rating: ", end = ""))) #I wrote this to make
print(a)

# if-elif statement is categorized by probability.
if (a == 4):
    print("\nProbability Rating: " + "4")
    print("Percent: " + "90-100%")
    print ("Description: " + "Almost Certain to certain")
elif (a == 3):
    print("\nProbability Rating: " + "3")
    print("Percent: " + "50-90%")
    print ("Description: " + "Likely")
elif (a == 2):
    print("\nProbability Rating: " + "2")
    print("Percent: " + "10-50%")
    print ("Description: " + "Possible")
elif (a == 1):
    print("\nProbability Rating: " + "1")
    print("Percent: " + "1-10%")
    print ("Description: " + "Rare")
elif (a == 0):
    print("\nProbability Rating: " + "0")
    print("Percent: " + "0%")
    print ("Description: " + "Item/operation is not present in laboratory.")
ov = a
```

Enter the probability rating: 4

Probability Rating: 4

Percent: 90-100%

Description: Almost Certain to certain

Step 2: Choice of Severity Scale

We don't have to choose the type of scale. This is because the type of scale can be determined by the given integer value. If the value is 1, it corresponds to "No Risk" whether it is WVS or SLS, a value of 4 or less is determined as SLS, and a value of 5 or more is determined as WVS.

```
In [26]: b = int(input(print("Enter the Severity: ", end = "")))
print(b)

if b <= 4:
    type_scale = "SLS"
else:
    type_scale = "WVS"
```

Enter the Severity: 4

Step 3: Hazard Severity

1. Thinking: CV values will be calculated according to four types of risk classification. At this time, the type of scale determined in the previous cell will also be considered.
2. Declare functions separately to reduce unnecessary code.
3. The function returns CV values and print information according to the entered risk classification.
4. Since it is inefficient to apply all print statements to each if statement, it will be effectively expressed using functions and lists.

```
In [27]: personal_saftey = ["No injuries", "Minor injuries", "Moderate to life impacting injuri
resources = ["No Impact", "Moderate Impact", "Additional resources required", "Institut
work_performane = ["No Delays", "Modest Delays", "Significant delays", "Major operatic
property_damage = ["Minor", "Moderate", "Substantial", "Severe"]
reputation = ["No impact", "Potential damage", "Damaged", "Loss of confidence"]

severity_info_list=[personal_saftey, resources, work_performane, property_damage, repu
severity_info_list_str=["personal saftey: ", "resources: ", "work performane: ", "pro
scale_rating = ["No Risk", "Minor", "Moderate", "High"]

if type_scale == "SLS":
    if b == 1:
        idx = 0
    elif b == 2:
        idx = 1
    elif b == 3:
        idx = 2
    elif b == 4:
        idx = 3
```

```

elif type_scale == "WVS":
    if b == 1:
        idx = 0
    elif b == 5:
        idx = 1
    elif b == 10:
        idx = 2
    elif b == 20:
        idx = 3

cv = b

severity_rating = scale_rating[idx]
print("You chose '%s,' The information is as follows:" %severity_rating)

for i in range (0, 5):
    print(severity_info_list_str[i] + severity_info_list[i][idx])

print("CV = %d" %cv)

```

You chose 'High,' The information is as follows:
 personal safety: Life threatening injuries from single exposure
 resources: Institutional resources required
 work performance: Major operational disruptions
 property damage: Severe
 reputation: Loss of confidence
 CV = 4

Step 4: Calculating and Summarizing the Risk Rating

1. Through the code above, we were able to get cv and ov.
2. The risk_rating is calculated according to this formula; Risk Rating (RR) = Probability of Occurrence (OV) x Severity of Consequences Value (CV)

```

In [28]: risk_rating = ov * cv
print("cv = %d, ov = %d"%(cv, ov))
print("Therefore, the riskrating can be calculated by multiplying two elements (cv * o

cv = 4, ov = 4
Therefore, the risk rating can be calculated by multiplying two elements (cv * ov): 16

```

Step 5: Allowing for multiple calculations

We can re-execute the code from start to finish using while statement. Put the termination condition at the end to prevent it from running again indefinitely. Also, declarations of lists that do not need to be looped do not need to go into while statement.

```

In [30]: personal_safety = ["No injuries", "Minor injuries", "Moderate to life impacting injuri
resources = ["No Impact", "Moderate Impact", "Additional resources required", "Institut
work_performane = ["No Delays", "Modest Delays", "Significant delays", "Major operatic
property_damage = ["Minor", "Moderate", "Substantial", "Severe"]
reputation = ["No impact", "Potential damage", "Damaged", "Loss of confidence"]

```

```

severity_info_list=[personal_saftey, resources, work_performane, property_damage, repu
severity_info_list_str=["personal saftey: ", "resources: ", "work performane: ", "pro
scale_rating = ["No Risk", "Minor", "Moderate", "High"]
SLS_cv = [1, 2, 3, 4]
WVS_cv = [1, 5, 10, 20]

while True:
    a = int(input(print("Enter the probability rating: ", end = ""))) #I wrote this to
    print(a)

    # if-elif statement is categorized by probability.
    if (a == 4):
        print("\nProbability Rating: " + "4")
        print("Percent: " + "90-100%")
        print ("Description: " + "Almost Certain to certain")
    elif (a == 3):
        print("\nProbability Rating: " + "3")
        print("Percent: " + "50-90%")
        print ("Description: " + "Likely")
    elif (a == 2):
        print("\nProbability Rating: " + "2")
        print("Percent: " + "10-50%")
        print ("Description: " + "Possible")
    elif (a == 1):
        print("\nProbability Rating: " + "1")
        print("Percent: " + "1-10%")
        print ("Description: " + "Rare")
    elif (a == 0):
        print("\nProbability Rating: " + "0")
        print("Percent: " + "0%")
        print ("Description: " + "Item/operation is not present in laboratory.")

    print("Please select the scale you want to use.")
    print("1. Standard Linear Scale")
    print("2. Weighted Value Scale")

    b = int(input(print("Enter the Severity: ", end = "")))
    print(b)

    if b <= 4:
        type_scale = "SLS"
    else:
        type_scale = "WVS"

    if type_scale == "SLS":
        if b == 1:
            idx = 0
        elif b == 2:
            idx = 1
        elif b == 3:
            idx = 2
        elif b == 4:
            idx = 3

    elif type_scale == "WVS":

```

```
    if b == 1:
        idx = 0
    elif b == 5:
        idx = 1
    elif b == 10:
        idx = 2
    elif b == 20:
        idx = 3

cv = b

severity_rating = scale_rating[idx]
print("You chose '%s,' The information is as follows:" %severity_rating)

for i in range (0, 5):
    print(severity_info_list_str[i] + severity_info_list[i][idx])

print("CV = %d" %cv)

risk_rating = ov * cv
print("cv = %d, ov = %d"%(cv, ov))
print("Therefore, the risk rating can be calculated by multiplying two elements (cv
print("If you don't want any more calculations, please enter 'no'.")

repeat = input()
if repeat == "no":
    break
```

Enter the probability rating: 4

Probability Rating: 4

Percent: 90-100%

Description: Almost Certain to certain

Enter the Severity: 4

You chose 'High,' The information is as follows:

personal safety: Life threatening injuries from single exposure

resources: Institutional resources required

work performance: Major operational disruptions

property damage: Severe

reputation: Loss of confidence

CV = 4

cv = 4, ov = 4

Therefore, the risk rating can be calculated by multiplying two elements (cv * ov): 16

If you don't want any more calculations, please enter 'no'.

Enter the probability rating: 2

Probability Rating: 2

Percent: 10-50%

Description: Possible

Enter the Severity: 20

You chose 'High,' The information is as follows:

personal safety: Life threatening injuries from single exposure

resources: Institutional resources required

work performance: Major operational disruptions

property damage: Severe

reputation: Loss of confidence

CV = 20

cv = 20, ov = 4

Therefore, the risk rating can be calculated by multiplying two elements (cv * ov): 80

If you don't want any more calculations, please enter 'no'.

Step 6: Summary

I wrote a program that calculates risk. The user enters a value and calculates the risk according to the prompts.

The weakness of this code is that when you enter an unexpected value, you get an incorrect result. In the case of exceptions, you can supplement this by coding additional numbers.

Since there is no function, so, it takes considerable effort to loop all the existing code through the while statement.

If all the code had been functionalized as I can and organized neatly from the beginning, Step 5 would have been easy.

Section 2: Data Validation

The parts that accept the input value in my code are probability rating and severity rating.

First, determine whether the accepted value is an integer, and then, secondly, check whether the integer value is a valid rating.

If it is not a valid input, an error message is output and an input value is received again.

Here, I'm going to make a function to check if input is a valid value. I am going to make a completed version in section 3.

```
In [2]: # If the return value is false, the loop ask again.

def valid_probability(x):
    try:
        x = int(x)
    except ValueError:
        print("Please enter a valid integer. [0, 1, 2, 3, 4]")
        return False

    if x in [0, 1, 2, 3, 4]:
        return True
    else:
        print("Please enter a valid integer. [0, 1, 2, 3, 4]")
        return False

def valid_Severity(x):
    try:
        x = int(x)
    except ValueError:
        print("Please enter a valid integer. [1, 2, 3, 4, 5, 10, 20]")
        return False

    if x in [1, 2, 3, 4, 5, 10, 20]:
        return True
    else:
        print("Please enter a valid integer. [1, 2, 3, 4, 5, 10, 20]")
        return False
```

The process of checking that the input value is a valid value is one of the important steps. This work reduces the scope of errors. Writing a function prevents unnecessary duplicate code writing and can be used more usefully.

Section 3: Functions and Input Validation

I functioned the code in Section 1 and added a function to validate the value. This reduces unnecessary duplicate code and makes it look better. In addition, by checking the valid values, systematic errors are prevented for input of invalid values.

```
In [4]: range_Probability = ["0%", "1-10%", "10-50%", "50-90%", "90-100%"]
description_Probability = ["Item/operation is not present in laboratory.", "Rare", "Po

personal_saftey = ["No injuries", "Minor injuries", "Moderate to life impacting injuri
resources = ["No Impact", "Moderate Impact", "Additional resources required", "Institut
work_performane = ["No Delays", "Modest Delays", "Significant delays", "Major operatic
property_damage = ["Minor", "Moderate", "Substantial", "Severe"]
reputation = ["No impact", "Potential damage", "Damaged", "Loss of confidence"]

severity_info_list=[personal_saftey, resources, work_performane, property_damage, repu
```

```

severity_info_list_str = ["personal saftey: ", "resources: ", "work performane: ", "proc

valid = False

SLS_cv = [1, 2, 3, 4]
WVS_cv = [1, 5, 10, 20]

# A function that provides a description based on the entered probability
def info_probability(x):
    print("Probability Rating: %d" %x)
    print("Percent: " + range_Probability[x])
    print ("Description: " + description_Probability[x])
    return

# A function that determine the type of scale based on the severity entered
# Instead of returning the scale type, it returns the index number for the code it des
def info_scale(x):
    if x <= 4:
        type_scale = "SLS"
    else:
        type_scale = "WVS"

    if type_scale == "SLS":
        if x == 1:
            idx = 0
        elif x == 2:
            idx = 1
        elif x == 3:
            idx = 2
        elif x == 4:
            idx = 3

    elif type_scale == "WVS":
        if x == 1:
            idx = 0
        elif x == 5:
            idx = 1
        elif x == 10:
            idx = 2
        elif x == 20:
            idx = 3

    return idx

# A function that provides a description based on the severity entered
def info_severity(x):
    for i in range (0, 5):
        print(severity_info_list_str[i] + severity_info_list[i][x])

# A function that returns risk rating by multiplying the entered cv and ov values
def get_risk_rating(x, y):
    return x * y

while True:

    while True:

```



```
a = input(print("Enter the probability rating: ", end = "")) #I wrote this to
print(a)
valid = valid_probability(a)
if valid:
    a = int (a)
    break

info_probability(a)
print("-----")
while True:
    b = input(print("Enter the Severity: ", end = ""))
    print(b)
    valid = valid_Severity(b)
    if valid:
        b = int(b)
        break

idx = info_scale(b)
info_severity(idx)
print("-----")
risk_rating = get_risk_rating(a, b)
print("The risk rating can be calculated by multipling two elements (cv * ov): %d"
print("-----")
print("If you don't want any more calculations, please enter 'no'.")
repeat = input()
if repeat == "no":
    print("The program is terminated.")
    break
print("-----")
```

```

Enter the probability rating: x
Please enter a valid integer. [0, 1, 2, 3, 4]
Enter the probability rating: -1
Please enter a valid integer. [0, 1, 2, 3, 4]
Enter the probability rating: 4
Probability Rating: 4
Percent: 90-100%
Description: Almost Certain to certain
-----
Enter the Severity: x
Please enter a valid integer. [1, 2, 3, 4, 5, 10, 20]
Enter the Severity: 7
Please enter a valid integer. [1, 2, 3, 4, 5, 10, 20]
Enter the Severity: 20
personal saftey: Life threatening injuries from single exposure
resources: Institutional resources required
work performane: Major operational disruptions
property damage: Severe
reputation: Loss of confidence
-----
The risk rating can be calculated by multipling two elements (cv * ov): 80
-----
If you don't want any more calculations, please enter 'no'.
-----
Enter the probability rating: 2
Probability Rating: 2
Percent: 10-50%
Description: Possible
-----
Enter the Severity: 4
personal saftey: Life threatening injuries from single exposure
resources: Institutional resources required
work performane: Major operational disruptions
property damage: Severe
reputation: Loss of confidence
-----
The risk rating can be calculated by multipling two elements (cv * ov): 8
-----
If you don't want any more calculations, please enter 'no'.
The program is terminated.

```

We can repeat the code through the loop. However, it is important to note that an end condition that can exit the loop should be inserted. The process of checking that the input value is a valid value is one of the important steps. This work reduces the scope of errors. Writing a function prevents unnecessary duplicate code writing and can be used more usefully.